SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:
Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

QUESTIONS: 5 **Total Marks:25**

Annexure A

Comparative Analysis

1. Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.
2. Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.
3. Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.
4. Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.
5. In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Concepts	15%	Demonstrates complete and in-depth understanding of all concepts being compared	Good understanding with minor gaps	Basic understanding; some concepts unclear	Poor understanding of concepts
Comparison Depth	20%	Provides detailed, insightful, and meaningful comparisons across multiple aspects	Adequate comparison with relevant points	Limited comparison; lacks depth	Minimal or incorrect comparison
Criteria Selection	10%	Selects highly relevant and appropriate comparison parameters	Mostly relevant criteria	Some irrelevant or missing criteria	Poor or no criteria selection
Analytical Skills	15%	Strong analysis with logical reasoning and clear justification	Good analysis with some justification	Basic analysis with weak reasoning	No clear analysis
Organization & Structure	10%	Well-organized, clear, and logically structured	Generally organized	Somewhat disorganized	Poorly structured
Use of Examples / Evidence	10%	Uses strong, relevant examples or data to support comparison	Uses some examples	Limited examples	No supporting examples
Clarity of Presentation	10%	Very clear, concise, and easy to understand	Mostly clear	Somewhat unclear	Difficult to understand
Conclusion & Insights	10%	Insightful conclusion with strong justification	Clear conclusion	Basic conclusion	No or weak conclusion

1. Star Topology vs Bus Topology – Cost, Reliability and Performance

Star and bus topologies differ significantly in terms of cost, reliability, and performance. In star topology, all devices are connected individually to a central device such as a switch or hub. Therefore, it requires more cables and additional networking hardware. This makes star topology more expensive to install compared to bus topology.

In bus topology, all devices share a single backbone cable. Since only one main cable is required and no central switch is necessary, the installation cost is relatively low. It is therefore suitable for small networks where cost is an important consideration.
In terms of reliability, star topology is better because failure of one device or its connecting cable does not affect the remaining devices. However, if the central switch or hub fails, the entire network can become unavailable. In bus topology, failure of the backbone cable can affect communication across the whole network.

Star topology also provides better performance because the central switch can manage data transmission efficiently. Each device generally has a dedicated connection to the switch. Bus topology uses a shared communication medium, so collisions and network congestion can increase when many devices transmit data simultaneously.

Therefore, star topology is preferred for modern networks because it provides better performance, reliability, and easier troubleshooting, whereas bus topology is mainly advantageous because of its low cost and simple structure.

2. Mesh Topology vs Star Topology – Fault Tolerance and Reliability

Mesh topology and star topology can be compared mainly based on their fault tolerance and reliability. In a mesh topology, devices are connected through multiple links, creating several possible paths for data transmission. This provides a high level of redundancy.

If one communication link fails in a mesh network, data can be transmitted through another available path. As a result, a single link failure normally does not interrupt the entire network. This makes mesh topology highly suitable for critical communication systems, data centers, military networks, and other environments where continuous connectivity is important.

Star topology provides moderate fault tolerance. If one computer or its cable fails, the other computers can continue communicating normally. However, the central switch or hub represents a single point of failure. If this central device fails, communication between the connected devices can be interrupted.

Mesh topology, however, requires a large number of cables, ports, and connections. This increases installation and maintenance costs. Star topology requires fewer connections and is easier to manage.

Therefore, mesh topology provides greater fault tolerance and reliability, while star topology provides a better balance between reliability, cost, and ease of management.

3. Bus Topology vs Mesh Topology – Installation Complexity

Bus and mesh topologies have completely different installation requirements. **Bus topology** uses a single backbone cable to which multiple devices are connected. Because devices share the same communication medium, fewer cables are required.
The simple structure of bus topology makes it relatively easy and inexpensive to install. It can be useful for small networks where the number of devices is limited. However, proper termination of the backbone cable is required, and locating faults in the backbone

can be difficult.

In **mesh topology**, devices are connected to multiple or all other devices. In a full mesh, every device has a direct connection with every other device. If there are many devices, the number of required connections increases rapidly.
For example, a full mesh network with **n devices requires n(n−1)/2 links**. Therefore, increasing the number of devices significantly increases installation complexity, cable requirements, and cost.

Although mesh topology is more difficult to install, it provides excellent reliability because multiple paths are available for communication.

Thus, bus topology is easier, cheaper, and faster to install, while mesh topology is more complex and expensive but provides superior reliability and fault tolerance.

4. Hybrid Topology vs Star Topology – Scalability

Scalability refers to the ability of a network to expand when additional devices or network segments are required. Hybrid topology provides greater flexibility because it combines two or more different topologies according to the requirements of an organization.

For example, an organization may use star topology within individual departments and connect these departments using another topology. This allows different sections of the network to be designed according to their specific requirements.
In star topology, additional devices can be connected to the central switch as long as free ports are available. If all ports are occupied, an additional switch may be required. Therefore, expansion is dependent on the capacity and design of the central networking device.

Hybrid topology provides greater flexibility because new segments can be added without completely redesigning the existing network. It is therefore suitable for **large organizations, universities, enterprises, and growing businesses**.

However, hybrid networks can be more expensive and require skilled administrators because different topology structures need to be managed together.

Therefore, hybrid topology provides greater scalability and flexibility, while star topology provides simpler and more straightforward expansion.

5. Star, Mesh, Bus and Hybrid Topologies – Maintenance

Maintenance is an important factor when selecting a network topology because network faults need to be detected and corrected quickly.

Star topology is generally easy to maintain. Since every device has a separate connection to the central switch, administrators can identify which device or cable is causing a problem. A failure in one connection normally affects only that particular device.

Mesh topology is more difficult to maintain because it contains many connections between devices. Although multiple links provide excellent reliability, troubleshooting and managing a large number of connections can be time-consuming. However, the presence of alternative paths makes the network highly resistant to individual link failures.

Bus topology is relatively simple in structure but can be difficult to troubleshoot. If the backbone cable develops a fault, multiple or all devices may lose communication. Finding the exact location of a fault in the shared backbone can take considerable time.

Hybrid topology has moderate to high maintenance requirements. Since it combines different topologies, administrators need to understand and monitor multiple network structures. A problem in one section may require checking several connected components.

Therefore, star topology is generally the easiest to maintain, bus topology is simple but difficult to troubleshoot when backbone failures occur, mesh topology is reliable but complex, and hybrid topology provides flexibility but requires skilled network administration.